

# SOMA: EFFICIENT MULTI-TURN LLM SERVING VIA SMALL LANGUAGE MODEL

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## ABSTRACT

Large Language Models (LLMs) are increasingly deployed in multi-turn dialogue settings where preserving conversational context across turns is essential. A standard serving practice concatenates the full dialogue history at every turn, which reliably maintains coherence but incurs substantial cost in latency, memory, and API expenditure, especially when queries are routed to large proprietary models. Existing approaches often struggle to balance the trade-off between response quality and efficiency. We propose a framework that exploits the early turns of a session to estimate a local response manifold and then adapt a smaller surrogate model to this local region for the remainder of the conversation. Concretely, we learn soft prompts that maximize semantic divergence between the large and surrogate small language models’ responses to surface least-aligned local directions, stabilize training with anti-degeneration control, and distill the mined cases into localized LoRA fine-tuning so the surrogate runs without prompts at inference. A simple gate enables a one-time switch with rollback on drift. We further provide a theoretical analysis for key components in SOMA. **Extensive experiments on six multi-turn benchmarks show that SOMA preserves the original model’s responses and reduces serving cost for medium-to-long, locally coherent sessions.** The source code is provided at: <https://anonymous.4open.science/r/SOMA-4175>.

## 1 INTRODUCTION

Large language models (LLMs) such as the GPT series (Radford et al., 2019; Brown et al., 2020; Achiam et al., 2023), LLaMA (Touvron et al., 2023), Claude (Anthropic, 2023), and DeepSeek (Guo et al., 2025) have demonstrated strong performance in real-world machine-learning-as-a-service (MLaaS) applications, ranging from chat assistants to code generation (Park & Kulkarni, 2023; Dong et al., 2023; Liu et al., 2024; Li et al., 2026; Yu et al., 2026). As LLMs are increasingly deployed interactively, *multi-turn LLM serving*, which covers extended interactions between humans and LLMs or among LLM agents, has become a key research focus (Yi et al., 2024; Li et al., 2025b; Zhao et al., 2025). However, supporting efficient context-dependent multi-turn interaction remains a key challenge, as most LLM serving systems are stateless and require resending the entire conversation history, including all prior queries and responses, with each new query to generate a new response (Ananda, 2025; Moon, 2025). This leads to redundant computation, high latency, and rising serving costs as conversations lengthen.

Previous work has explored efficient multi-turn LLM serving through two main approaches. One line of work focuses on *single-model methods* that compress dialogue history (Wang et al., 2025; Chen et al., 2024b; Xiao et al., 2024), retrieve memory from external modules (Melz, 2023; Gutiérrez et al., 2024), or reuse attention computations (Gao et al., 2024; Jeong & Ahn, 2025; Anthropic, 2024). However, these still rely heavily on large LLMs for every turn, leading to high monetary cost, latency, and GPU usage. They also often truncate or overlook extended context, limiting reasoning over long dialogues. Another line adopts *multi-model methods*, routing simple queries to smaller models while escalating harder ones to larger LLMs (Behera et al., 2025; Schick et al., 2023; Ding et al., 2024; Shnitzer et al., 2023). Yet, small models struggle to generalize across dialogue complexity, and model switching introduces additional overhead. Moreover, LLMs are known to rely on early turns (Xiao et al., 2024; Laban et al., 2025), compounding the difficulty of maintaining coherence.

*Is it possible to achieve an efficient, context-aware multi-turn LLM serving framework that avoids recomputing the full history at every turn while preserving response quality?*

To answer this question, we examine real-world multi-turn dialogues across domains and find a long-tail distribution of token counts over turns: early turns are substantially longer, while later turns shrink. This phenomenon aligns with the intuition of prior work that early turns typically carry **the substantive openings that set the issues and anchors of a conversation**, including questions, requests, and proposals, while later turns are typically minimal acknowledgments (He et al., 2018; Stolcke et al., 2000). This long-tail trend gives rise to an intuitive idea: since later turns are relatively short, a smaller language model might suffice to generate responses more cost-efficiently. The bottleneck, however, is the “big head”: most grounding is established early, and later turns remain highly dependent on it. A small model may behave reasonably at the start, but its responses drift from the large model as the dialogue progresses, so simply handing later turns to a small model degrades quality. The small model must therefore not only process shorter inputs but also approximate the larger model’s behavior within the local region of its output space that the prior dialogue induces. We call this a local manifold approximation problem.

Building on these insights, we present SOMA (Soft-prompts for lOcal Manifold Approximation), which adapts a small language model to the local behavior of a large one, conditioned on the early turns of a session, through a three-stage pipeline: (1) *soft-prompt tuning*, which explores the local region induced by the early context to find the directions of maximal behavioral divergence between the two models; (2) *localized fine-tuning*, which aligns the small model with the large one on a handful of mined input–output pairs; and (3) *efficient inference*, which serves later turns from a compressed context and rolls back to the large model when the topic drifts. Our contributions are:

- **Long-tail pattern in multi-turn dialogues.** We reveal an under-explored long-tail pattern in multi-turn dialogues: the first few turns concentrate heavy context, while later turns are shorter yet more dependent on earlier ones. This suggests large savings if a small, cheap model can serve the later turns once it is given the accumulated context.
- **SOMA: Efficient multi-turn serving.** It first learns soft prompts that expose the largest surrogate–original response dissimilarity, then adapts the surrogate localized LoRA accordingly, enabling prompt-free inference with a simple cosine gate for switching and rollback.
- **Theory analysis and empirical evaluation.** We provide concentration-based bounds for switching, coverage guarantees for prompt-direction search, and suboptimality limits for selected directions. Guided by these results, empirical studies show the effectiveness of SOMA in real world.

## 2 PRELIMINARIES

### 2.1 NOTATIONS

A multi-turn dialogue prefix of length  $k$  is  $\mathcal{D}_k = \{(q_1, a_1), \dots, (q_k, a_k)\}$ , where  $q_t$  is the user query at turn  $t$  and  $a_t$  is the model response. We write  $F$  for the original model, a proprietary black-box LLM as is common in machine-learning-as-a-service (MLaaS), and  $G$  for the small surrogate. The textual response of model  $M \in \{F, G\}$  at turn  $t$  is  $a_t^M$ . Let  $f_M(\cdot)$  map an input sequence to the hidden space, so that  $\mathbf{h}_t = f_M(q_{\leq t}) \in \mathbb{R}^d$  is the hidden state at turn  $t$ ; the first  $k$  hidden states  $\mathcal{H}_k = \{\mathbf{h}_1, \dots, \mathbf{h}_k\}$  induce a local region  $\mathcal{M}_k^M \subset \mathbb{R}^d$ . A soft prompt of length  $L$  is a matrix  $\mathbf{P} \in \mathbb{R}^{L \times d}$ . Appendix A lists all symbols.

### 2.2 EXPLORING TOKEN–TURN PATTERNS IN MULTI-TURN LLM DIALOGUES

Standard serving processes each new request together with the entire history, so the cost of a turn grows with the length of the conversation. The amount of *new* information in a turn, however, need not grow at all. We therefore first study token usage across turns.

We consider four representative settings: *ShareGPT* (Chen et al., 2024a) for open-ended human–LLM chat, *ReMeDi* (Yan et al., 2022) for doctor–patient consultation, *Craigslist Bargain* (He et al., 2018) for negotiation, and *Multi-Character* (agentlans, 2024) for multi-party role-play. Together they cover different speaker roles, task goals, and interaction structures. For each dataset, we compute the average token count at turn  $t$  and normalize it by the average of Turn 1.

Figure 1 shows a consistent long-tail pattern. Early turns are token-heavy: they introduce the task, the topic, the constraints, and the speaker’s intent. Later turns are much shorter and settle at a low, stable level. Shorter does not mean independent. Many later turns read as local updates to a dialogue state that has already been established, so the marginal new input shrinks while standard serving keeps re-processing the full prefix. This mismatch motivates an adaptive strategy: let the original model establish the early dialogue state, and hand subsequent turns to a cheaper surrogate for as long as they stay within that state, instead of recomputing the full history.

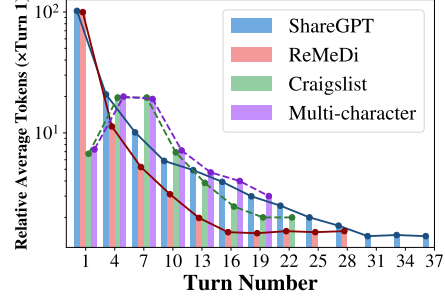


Figure 1: Average token count per turn, normalized by Turn 1, on four dialogue datasets: usage drops after the early turns and forms a long tail.

### 2.3 LOCAL MANIFOLD APPROXIMATION FOR MULTI-TURN DIALOGUES

The long-tail pattern creates an opportunity for surrogate serving, but a naive substitution is unreliable. A later turn is short precisely because much of its meaning is supplied by the preceding dialogue, so a surrogate that is not aligned with the original model under that prefix produces fluent responses that quietly drift from the intended context.

We describe this challenge through a local manifold view. Given a prefix  $\mathcal{D}_k$ , the original model  $F$  and the surrogate  $G$  each induce a local region in their contextual representation space, and this region captures the current topic, constraints, style, and task state. The goal is to make the surrogate’s region match that of the original model under the same prefix.

**Problem 2.1** (Local Manifold Approximation for Multi-Turn Interactions). *Given a dialogue prefix  $\mathcal{D}_k$  and an original model  $F$ , learn a surrogate model  $G \in \mathcal{G}$  whose local representation region matches that of  $F$  under the same prefix:*

$$\min_{G \in \mathcal{G}} \text{dist}(\mathcal{M}_k^G(\mathcal{D}_k), \mathcal{M}_k^F(\mathcal{D}_k)),$$

where  $\mathcal{M}_k^F(\mathcal{D}_k)$  and  $\mathcal{M}_k^G(\mathcal{D}_k)$  denote the local representation regions induced by the prefix, and  $\text{dist}(\cdot, \cdot)$  measures their discrepancy.

This formulation leads directly to SOMA. The early turns are used to expose where the surrogate disagrees with the original model within the current dialogue state, and these locally informative examples are then used to adapt the surrogate. After adaptation, the surrogate serves later turns only while the dialogue remains close to the learned region; when the conversation drifts, the system returns to the original model and refreshes the local state.

## 3 SOMA: LOCAL MANIFOLD APPROXIMATION VIA SOFT PROMPTS

SOMA has three stages. The original model  $F$  serves the early turns and thereby establishes the dialogue state. Soft-prompt mining then finds where the surrogate  $G$  is least aligned with  $F$  within that state, and the hard cases it exposes are distilled into a lightweight LoRA adapter. Finally, a semantic gate hands the session to  $G$  once local alignment is sufficient, and a drift detector returns it to  $F$  when the dialogue leaves the learned region.

### 3.1 INITIALIZATION

We initialize a soft prompt  $\mathbf{P} \in \mathbb{R}^{L_p \times d}$  on the surrogate  $G$  by sampling each row i.i.d. from  $\mathbf{p}_\ell \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ , where  $\sigma > 0$  sets the initialization scale. Let  $\mathbf{E} = [\mathbf{e}_v]_{v \in \mathcal{V}} \in \mathbb{R}^{d \times |\mathcal{V}|}$  be the surrogate embedding matrix and let  $\text{tok}(v)$  denote the token string associated with index  $v$ .

Because the original model  $F$  cannot accept continuous embedding vectors, the soft prompt is used in two aligned forms. The surrogate receives  $\mathbf{P}$  directly at the embedding layer, while the original model receives a nearest-neighbor verbalization of the same prompt:

$$v_\ell = \arg \max_{v \in \mathcal{V}} \frac{\langle \mathbf{p}_\ell, \mathbf{e}_v \rangle}{\|\mathbf{p}_\ell\|_2 \|\mathbf{e}_v\|_2}, \quad V(\mathbf{P}) = (\text{tok}(v_1), \dots, \text{tok}(v_{L_p})).$$

Let  $\mathcal{D}_{t-1}$  denote the text history before turn  $t$  and let  $q_t$  be the user query. During prompt mining, the two models are probed under comparable text conditioning:

$$a_t^F = F(V(\mathbf{P}) \oplus \mathcal{D}_{t-1} \oplus q_t), \quad a_t^G = G(\mathbf{P} \oplus_{\text{emb}} \mathbf{E}(\text{tok}_G(\mathcal{D}_{t-1} \oplus q_t))).$$

Here  $\text{tok}_G(\cdot)$  is the tokenizer of  $G$ ,  $\mathbf{E}(\cdot)$  maps text tokens to embeddings, and  $\oplus_{\text{emb}}$  concatenates along the sequence dimension. This construction lets  $F$  and  $G$  be compared under the same dialogue state while allowing gradients to flow only through  $\mathbf{P}$  on  $G$ ; the original-model responses used as mining targets are generated once in the warm-start window and kept fixed during optimization, so  $F$  is never re-queried while  $\mathbf{P}$  changes (Appendix G).

### 3.2 SOFT PROMPT TUNING FOR MINING WEAK ALIGNMENT DIRECTIONS

We learn  $\mathbf{P}$  by minimizing a differentiable objective that pushes  $G$  away from  $F$  on the local dialogue context. The objective has three parts: a token-level semantic divergence loss, an expectation-weighted distribution-level term, and an anti-degeneration regularizer.

**Semantic divergence loss.** We use an unlikelihood-style loss (Welleck et al., 2019) to discourage  $G$  from assigning high probability to the tokens produced by  $F$ . A purely exact-token penalty is too narrow, since the surrogate could avoid the teacher token while still assigning high probability to close paraphrases. We therefore define a semantic neighborhood in the surrogate embedding space.

**Definition 3.1** (Semantic Neighborhood). For token  $u \in \mathcal{V}$ , let  $\mathcal{N}_k(u)$  be the set of the  $k$  tokens in  $\mathcal{V} \setminus \{u\}$  whose embeddings have the largest cosine similarity to  $e_u$ .

At turn  $t$ , the original model produces text  $a_t^F$ . We tokenize this text using the surrogate tokenizer,  $y_t^F = \text{tok}_G(a_t^F) = (y_{t,1}^F, \dots, y_{t,T_F(t)}^F)$ , so that the teacher output and surrogate distribution are expressed in the same vocabulary. For each teacher token, we assign temperature-weighted mass to the token and its embedding neighbors:

$$s_\tau(v | y_{t,i}^F) = \frac{\exp(\cos(e_v, e_{y_{t,i}^F})/\tau)}{\sum_{u \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} \exp(\cos(e_u, e_{y_{t,i}^F})/\tau)}, \quad \tau > 0.$$

Let  $S_t = \mathcal{D}_{t-1} \oplus q_t$ . With causal masking, a single forward pass of  $G$  on the prefix  $S_t$  and the teacher answer prefix gives the next-token distributions  $\{\Pi_{t,i}(\mathbf{P})\}_{i=1}^{T_F(t)}$ , where  $\Pi_{t,i}(\mathbf{P}) = \pi_G(\cdot | S_t; \mathbf{P}, a_{t,<i}^F)$ . The semantic divergence loss is

$$\mathcal{L}_{\text{sem}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) = \frac{1}{T_F(t)} \sum_{i=1}^{T_F(t)} \sum_{v \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} s_\tau(v | y_{t,i}^F) \left[ -\log(1 - \Pi_{t,i}(\mathbf{P})[v]) \right]. \quad (1)$$

Minimizing Eq. 1 searches for prompts that expose local disagreement between  $F$  and  $G$  not only on exact tokens but also on nearby semantic alternatives.

**Expectation-weighted semantic divergence.** The token-neighborhood loss may still miss distribution-level alignment:  $G$  can reduce probability on a teacher token and its nearest neighbors while spreading mass over many other tokens with similar meaning. To detect this case, we compare the expected embedding of the surrogate’s next-token distribution with the embedding of the teacher token. Since  $\mathbf{E} \in \mathbb{R}^{d \times |\mathcal{V}|}$  stores token embeddings as columns, the expected embedding at position  $i$  is  $\bar{e}_{t,i} = \mathbf{E} \Pi_{t,i}(\mathbf{P}) = \sum_{v \in \mathcal{V}} \Pi_{t,i}(\mathbf{P})[v] e_v$ . We therefore upweight the neighborhood penalty whenever the expected embedding still points toward the teacher token, using the weight  $w_{t,i}(\mathbf{P}) = 1 + \lambda_{\text{exp}} \text{clip}(\cos(\bar{e}_{t,i}, e_{y_{t,i}^F}), 0, 1)$  with  $\lambda_{\text{exp}} \geq 0$ . The final semantic mining loss is

$$\mathcal{L}_{\text{sem\_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) = \frac{1}{T_F(t)} \sum_{i=1}^{T_F(t)} w_{t,i}(\mathbf{P}) \sum_{v \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} s_\tau(v | y_{t,i}^F) \left[ -\log(1 - \Pi_{t,i}(\mathbf{P})[v]) \right]. \quad (2)$$

Thus the loss blocks local copying through the neighborhood term and detects broader semantic shadowing through the expectation weight.

**Theorem 3.2** (Directional Recovery). *Let  $\mathbf{C} = \mathbb{E}[\mathbf{J}^\top \mathbf{J}]$  be the local discrepancy Fisher matrix of  $G$  over the warm-start window, where  $\mathbf{J}$  is the Jacobian of  $G$ ’s log next-token probabilities with respect to the soft prompt  $\mathbf{P}$ . If the local discrepancy is smooth and approximately rank  $r$ , then optimizing Eq. 2 recovers prompt directions whose span covers at least a  $(1 - \varepsilon)$  fraction of the top- $r$  eigenmass of  $\mathbf{C}$ , with smaller  $\varepsilon$  under better warm-start and neighborhood coverage.*

**Anti-degeneration regularizer.** Soft-prompt mining can collapse if  $G$  assigns excessive probability to a few high-frequency tokens, producing repetitive or bland continuations that are uninformative about where  $F$  and  $G$  differ (Li et al., 2023; Holtzman et al., 2019; Meister et al., 2023). We therefore preserve entropy near the prompt-context boundary. Reusing the logits from the same forward pass, define

$$H_{\text{tail}}(\mathbf{P}; t) = \frac{1}{K} \sum_{i \in \text{tail}(t, K)} \left[ - \sum_{v \in \mathcal{V}} \Pi_{t,i}(\mathbf{P})[v] \log \Pi_{t,i}(\mathbf{P})[v] \right].$$

The complete prompt-mining objective for a minibatch of warm-start turns  $\mathcal{B}$  is

$$\mathcal{J}(\mathbf{P}) = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} [\mathcal{L}_{\text{sem\_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) - \beta H_{\text{tail}}(\mathbf{P}; t)] + \lambda_P \|\mathbf{P}\|_F^2. \quad (3)$$

We minimize Eq. 3 with AdamW while all weights of  $G$  stay frozen, and we clip gradients to stabilize the prompt scale. Two choices keep mining cheap: semantic neighbors are retrieved from an ANN index over normalized token embeddings, and  $\bar{e}_{t,i}$  uses a top- $m$  truncation of  $\Pi_{t,i}(\mathbf{P})$  instead of a dense sum over the vocabulary. The per-step overhead is therefore linear in the number of teacher-prefix positions and scales with  $k + m$  rather than  $|\mathcal{V}|$ .

### 3.3 LOCALIZED FINE-TUNING, SWITCHING, AND ROLLBACK

The learned soft prompts are session-local mining tools, not inference-time inputs. After mining, SOMA converts them into ordinary local supervision. Each warm-start turn receives a hardness score equal to its largest mined divergence,  $r_t = \max_{\mathbf{P} \in \mathcal{P}_{\text{cand}}} \mathcal{L}_{\text{sem\_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t)$ , and the highest-scoring turns together with their original-model responses form a local training set  $\mathcal{H}$ . **This makes the data-mining role of soft prompts explicit: instead of training on all or randomly chosen warm-start turns, SOMA trains on the turns that prompt mining exposes as least aligned.**

We freeze the base weights of  $G$ , attach LoRA (Hu et al., 2022) to attention and MLP projections, and optimize

$$\mathcal{L}_{\text{FT}}(\boldsymbol{\Theta}_{\text{LoRA}}) = \frac{1}{|\mathcal{H}|} \sum_{(S_t, a_t^F) \in \mathcal{H}} \omega_t \text{NLL}(a_t^F | G(S_t; \boldsymbol{\Theta}_{\text{LoRA}})), \quad \omega_t \propto r_t. \quad (4)$$

The continuous soft prompts are discarded after this stage; once SOMA switches, the online serving path uses only the LoRA-adapted surrogate and a compressed text context.

Before switching, SOMA verifies two conditions: the adapted surrogate must reproduce the original model’s responses on a small acceptance batch, and the current query must remain close to the warm-start dialogue centroid. Once accepted,  $G$  serves later turns from a compressed input  $\tilde{S}_t$  that keeps a fixed-budget summary of earlier content together with the most recent  $K$  turns, so the full history is never re-processed. During surrogate serving, SOMA keeps monitoring semantic drift. If recent queries move away from the learned local state, it rolls back to  $F$ , refreshes the summary and the centroid, and returns to  $G$  only after the acceptance gate passes again. **Because the drift statistic is a continuous distance to the warm-start centroid, gradual drift lowers it monotonically and triggers rollback once the margin is exhausted, rather than only on abrupt shifts.**

## 4 THEORETICAL ANALYSIS

This section answers three design questions: when SOMA may accept the session-local surrogate, when its warm-start overhead is amortized, and how many soft-prompt candidates are needed. **SOMA is effective because later turns remain tied to this warm-start state, not because later turns are intrinsically easier than early ones. The results characterize operating conditions; they do not claim an advantage over alternative mining strategies, which we assess empirically in Section 5. Full statements and proofs are given in Appendix E.**

#### 4.1 WHEN TO SWITCH

At turn  $t$ , let  $S_t = \mathcal{D}_{t-1} \oplus q_t$  be the shared context, and let  $a_t^F$  and  $a_t^G$  be the responses of the original model and of the adapted surrogate. We measure their discrepancy with the bounded score  $\text{Gap}_\Theta(S_t) = 1 - \cos(\phi(a_t^G), \phi(a_t^F)) \in [0, 1]$ , where  $\phi$  is a fixed sentence encoder and  $\Theta$  is the current adapter. Because consecutive turns are dependent, all bounds use effective sample sizes,  $W_{\text{eff}}$  for the warm-start window and  $|B|_{\text{eff}}$  for an acceptance batch, which shrink when the early turns are noisy or topic-mixed. Lemma E.1 shows that any bounded warm-start statistic concentrates around its mean at rate  $\sqrt{2 \log(2/\delta)/W_{\text{eff}}}$ , which is why SOMA demands a longer warm start when the early turns do not define a coherent local state.

The switching decision compares the surrogate before and after adaptation. On an acceptance batch  $B$ , let  $X_j = \text{Gap}_{\Theta_{\text{old}}}(S_j) - \text{Gap}_{\Theta_{\text{new}}}(S_j)$  and  $\hat{\Delta}_B = \frac{1}{|B|} \sum_{S_j \in B} X_j$ , so that a positive  $\hat{\Delta}_B$  means that adaptation reduced the gap on the batch.

**Theorem 4.1** (Acceptance Bound for Switching). *Assume  $X_j \in [-1, 1]$  and let  $\gamma_B(\delta) = \sqrt{2 \log(1/\delta)/|B|_{\text{eff}}}$ . With probability at least  $1 - \delta$ ,  $\mathbb{E}[X] \geq \hat{\Delta}_B - \gamma_B(\delta)$ . Hence, if  $\hat{\Delta}_B \geq \varepsilon + \gamma_B(\delta)$ , the true expected improvement is at least  $\varepsilon$  with confidence  $1 - \delta$ .*

Combining both bounds yields SOMA’s switching rule (Corollary E.2): choose  $W$  so that the warm-start uncertainty is at most  $\eta$  and  $|B|$  so that  $\gamma_B(\delta) \leq \gamma$ , and switch only when both the output-fidelity gate and the locality gate pass with margins larger than  $\eta + \gamma$ , which controls the decision error at confidence  $1 - 2\delta$ . Short but context-dependent later turns still pass the locality gate because they remain close to the established dialogue state, whereas an abrupt topic change shifts the locality statistic, fails the gate, and triggers an immediate rollback to the original model.

#### 4.2 WHEN DOES SOMA GIVE NET EFFICIENCY GAINS?

SOMA pays a one-time warm-start overhead, so its efficiency must be judged per session. Let  $T$  be the number of turns,  $W$  the switch point, and  $c_F(\cdot)$  and  $c_G(\cdot)$  the per-turn costs (latency, tokens, or price) of the original model and of the surrogate on its compressed input  $\tilde{S}_t$ . Relative to serving every turn with  $F$ , the net gain of SOMA is

$$\Delta C(T; W) = \sum_{t=W+1}^T \left[ c_F(\mathcal{D}_{t-1} \oplus q_t) - c_G(\tilde{S}_t) \right] - C_{\text{probe}}(W) - C_{\text{LoRA}}(W) - C_{\text{rb}}(T; W), \quad (5)$$

where  $C_{\text{probe}}$  and  $C_{\text{LoRA}}$  are the costs of soft-prompt mining and localized adaptation, and  $C_{\text{rb}}$  covers drift checks and occasional rollback. SOMA is therefore beneficial only when the accumulated post-switch savings exceed the one-time overhead. **This explains the operating range measured in Table 4: sessions of 1–4 turns do not benefit, whereas locally coherent sessions of 5 or more turns amortize the initialization cost, with the gain growing in session length.**

#### 4.3 HOW MANY SOFT-PROMPT CANDIDATES?

Near a reference prompt, assume that the mining objective is locally quadratic with curvature  $\mathbf{H}_T \succeq 0$  whose energy concentrates in an  $r_{\text{act}}$ -dimensional active subspace with dominant unit direction  $\mathbf{u}_1$ . SOMA draws  $M$  random candidates and keeps the one with the largest mining loss.

**Theorem 4.2** (Coverage of the Best Local Direction). *For  $r_{\text{act}} \geq 2$ , let  $\{\mathbf{u}_m\}_{m=1}^M$  be i.i.d. uniform unit vectors in the active subspace, and let  $p_{r_{\text{act}}}(\theta)$  be the probability that one of them lies within angle  $\theta \in (0, \pi/2]$  of  $\mathbf{u}_1$ . Then  $\Pr(\min_m \angle(\mathbf{u}_m, \mathbf{u}_1) \leq \theta) = 1 - (1 - p_{r_{\text{act}}}(\theta))^M$ ; when  $r_{\text{act}} = 1$ , a single candidate already suffices.*

Corollary E.4 turns this into a budget: coverage  $1 - \delta$  is guaranteed once  $M \geq \lceil \log \delta / \log(1 - p_{r_{\text{act}}}(\theta)) \rceil$ . A candidate within angle  $\theta$  of  $\mathbf{u}_1$  retains at least a  $\cos^2 \theta$  fraction of the top curvature (Lemma E.3), which turns the candidate count into an explicit quality trade-off. We set  $W$ ,  $|B|$ , and  $M$  from these bounds in all experiments (Appendix C.2).

Table 1: Response similarity to the original model on the LLaMA family. Higher is better.

|                | ShareGPT                          | ReMeDi                            | Craigslist                        | Multi-Char                        | MATH                              | MT-Bench                          | Avg.                              |
|----------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Surrogate      | 79.2 $\pm$ 2.18                   | 82.7 $\pm$ 1.95                   | 74.3 $\pm$ 2.36                   | 70.8 $\pm$ 1.73                   | 66.2 $\pm$ 2.91                   | 77.5 $\pm$ 2.04                   | 75.1 $\pm$ 5.98                   |
| History-Prefix | 86.1 $\pm$ 1.67                   | 87.9 $\pm$ 1.55                   | 82.4 $\pm$ 1.72                   | 84.7 $\pm$ 1.69                   | 80.3 $\pm$ 3.83                   | 87.2 $\pm$ 1.58                   | 84.8 $\pm$ 2.94                   |
| History-FT     | 93.4 $\pm$ 2.12                   | 91.8 $\pm$ 2.09                   | 90.3 $\pm$ 1.23                   | 89.1 $\pm$ 2.23                   | 87.6 $\pm$ 2.26                   | 92.4 $\pm$ 1.14                   | 90.8 $\pm$ 2.18                   |
| LLMLingua-2    | 84.6 $\pm$ 1.72                   | 86.4 $\pm$ 1.63                   | 80.8 $\pm$ 1.91                   | 82.9 $\pm$ 1.58                   | 78.1 $\pm$ 2.77                   | 85.3 $\pm$ 1.66                   | 83.0 $\pm$ 3.03                   |
| RouteLLM       | 95.3 $\pm$ 1.44                   | 92.5 $\pm$ 1.07                   | 91.0 $\pm$ 1.86                   | 91.4 $\pm$ 1.23                   | 89.6 $\pm$ 1.95                   | 93.2 $\pm$ 1.12                   | 92.2 $\pm$ 1.78                   |
| <b>SOMA</b>    | <b>96.4 <math>\pm</math> 1.91</b> | <b>93.2 <math>\pm</math> 0.98</b> | <b>91.9 <math>\pm</math> 2.49</b> | <b>92.3 <math>\pm</math> 1.05</b> | <b>90.7 <math>\pm</math> 1.12</b> | <b>94.1 <math>\pm</math> 0.91</b> | <b>93.1 <math>\pm</math> 1.99</b> |

## 5 EMPIRICAL STUDIES OF THE EFFECTIVENESS OF SOMA

We evaluate SOMA through five questions. **RQ1:** Can a small surrogate preserve the original model’s responses? **RQ2:** Does higher response similarity also improve task-grounded quality? **RQ3:** When does SOMA provide net efficiency gains after warm-start and adaptation overhead? **RQ4:** Which components of SOMA are most important? **RQ5:** When is session-local serving reliable, and when should SOMA fall back to the large model?

**Datasets.** We evaluate SOMA on six multi-turn benchmarks: ShareGPT (Chen et al., 2024a), ReMeDi (Yan et al., 2022), Craigslist (He et al., 2018), Multi-Char (agentlans, 2024), MATH (Hendrycks et al., 2021), and MT-Bench (Zheng et al., 2023). Dataset details and the context-dependence filtering are in Appendix C.1.

**Models and baselines.** We test two model families: LLaMA, using LLaMA-3.1-70B as  $F$  and LLaMA-2-7B as  $G$ , and Qwen, using Qwen-3-8B as  $F$  and Qwen-3-0.6B as  $G$ . We compare with Original, Surrogate, History-Prefix, History-FT, LLMLingua-2 (Pan et al., 2024), and RouteLLM (Ong et al., 2024); Appendix C.3 details each baseline.

**Metrics.** We mainly measure teacher-fidelity similarity, i.e., how closely each method preserves the original model’s response, using three LLM judges: GPT-OSS (OpenAI, 2025), DeepSeek-V3, and Gemma-2-27B. To complement similarity with task-grounded quality, we report EM accuracy on MATH. Efficiency is measured by token usage, throughput, and end-to-end wall-clock latency, which for SOMA includes soft-prompt mining, LoRA adaptation, and gating. All models are served with vLLM 0.10.2 with automatic prefix caching enabled, so every baseline already benefits from KV-cache reuse across the turns of a session.

**Implementation.** We set the warm-start window, acceptance batch, prompt candidates, and switch threshold according to the analysis in Section 4. After switching, SOMA discards soft prompts, performs no further LoRA updates, and only keeps the lightweight drift check in Section 3. More details, including hyperparameter ranges, are in Appendix C.2.

### 5.1 EXPERIMENTAL RESULTS

**RQ1: Response similarity to the original model.** After the warm-start stage, the surrogate should preserve the original model’s behavior within the current dialogue state. Table 1 reports response similarity on the LLaMA family. SOMA achieves the highest average similarity across all six datasets. The gap over Surrogate shows the behavior loss caused by directly replacing the large model with a smaller model. The gap over History-FT shows that local adaptation helps, but becomes stronger when SOMA selects training examples from weak-alignment directions. The largest gains appear on MATH and Multi-Char, where later turns depend on accumulated reasoning states, constraints, and roles. Table 2 repeats the evaluation on the Qwen family. Absolute similarities are lower because the capacity gap between Qwen-3-8B and Qwen-3-0.6B is wider, which makes local imitation harder, yet SOMA is again the best method on every dataset.

**RQ2: Task-grounded quality.** Response similarity is the primary metric for replacing the original model, but similarity alone does not guarantee task quality. We therefore evaluate MATH using exact-match accuracy. Table 3 shows that SOMA remains below the original large model, as expected, but substantially narrows the gap between the surrogate and the original model. SOMA also outperforms History-Prefix, History-FT, and RouteLLM in both model families. Thus SOMA’s gains are not only surface-level matching: local adaptation also preserves more of the original model’s reasoning.

Table 2: Response similarity to the original model on the Qwen family. Higher is better.

|                | ShareGPT                          | ReMeDi                            | Craigslist                        | Multi-Char                        | MATH                              | MT-Bench                          | Avg.                              |
|----------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Surrogate      | 50.9 $\pm$ 1.21                   | 63.5 $\pm$ 2.87                   | 48.2 $\pm$ 3.40                   | 44.7 $\pm$ 3.54                   | 42.6 $\pm$ 1.70                   | 40.4 $\pm$ 0.81                   | 48.4 $\pm$ 8.31                   |
| History-Prefix | 70.5 $\pm$ 1.15                   | 75.7 $\pm$ 2.01                   | 63.0 $\pm$ 2.23                   | 65.5 $\pm$ 0.91                   | 51.3 $\pm$ 2.45                   | 53.6 $\pm$ 2.32                   | 63.3 $\pm$ 9.47                   |
| History-FT     | 76.4 $\pm$ 1.68                   | 79.2 $\pm$ 2.61                   | 74.5 $\pm$ 3.56                   | 63.2 $\pm$ 1.74                   | 65.9 $\pm$ 1.70                   | 62.7 $\pm$ 0.72                   | 70.3 $\pm$ 7.23                   |
| LLMLingua-2    | 68.9 $\pm$ 1.32                   | 74.1 $\pm$ 2.14                   | 61.2 $\pm$ 2.67                   | 63.4 $\pm$ 1.18                   | 49.8 $\pm$ 2.03                   | 51.7 $\pm$ 1.45                   | 61.5 $\pm$ 7.92                   |
| RouteLLM       | 79.6 $\pm$ 1.04                   | 81.9 $\pm$ 1.36                   | 75.1 $\pm$ 2.91                   | 72.8 $\pm$ 1.62                   | 67.3 $\pm$ 1.21                   | 68.0 $\pm$ 0.96                   | 74.1 $\pm$ 5.43                   |
| <b>SOMA</b>    | <b>81.0 <math>\pm</math> 0.93</b> | <b>83.2 <math>\pm</math> 1.21</b> | <b>76.4 <math>\pm</math> 3.85</b> | <b>74.2 <math>\pm</math> 2.53</b> | <b>68.7 <math>\pm</math> 1.28</b> | <b>69.2 <math>\pm</math> 1.08</b> | <b>75.5 <math>\pm</math> 5.97</b> |

Table 3: MATH exact-match accuracy. SOMA improves task-grounded quality over baselines.

| Family | Orig.            | Surr.            | Hist.-P          | Hist.-FT         | Route            | SOMA                               |
|--------|------------------|------------------|------------------|------------------|------------------|------------------------------------|
| LLaMA  | 48.34 $\pm$ 0.32 | 19.20 $\pm$ 0.78 | 25.03 $\pm$ 0.94 | 31.46 $\pm$ 0.87 | 33.88 $\pm$ 0.79 | <b>41.62 <math>\pm</math> 0.66</b> |
| Qwen   | 36.48 $\pm$ 0.41 | 11.73 $\pm$ 0.64 | 16.21 $\pm$ 0.79 | 22.57 $\pm$ 0.83 | 25.08 $\pm$ 0.71 | <b>31.14 <math>\pm</math> 0.74</b> |

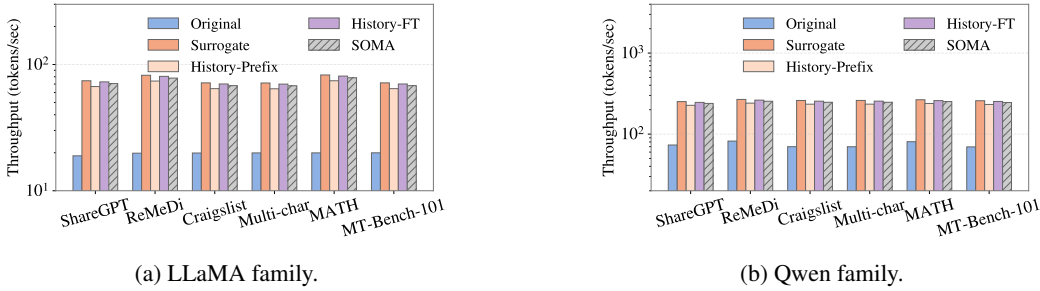


Figure 2: Throughput across six datasets. SOMA keeps the serving path close to the surrogate while maintaining higher similarity to the original model.

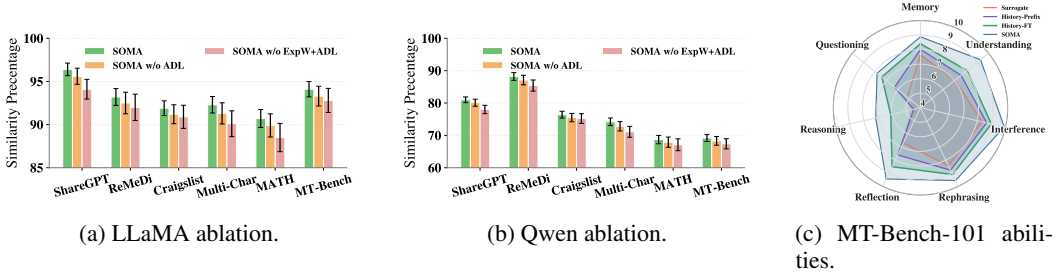


Figure 3: Component and capability analysis. Both anti-degeneration regularization and expectation-weighted divergence improve mining on LLaMA (a) and Qwen (b), and SOMA gains across MT-Bench-101 abilities (c).

**RQ3: End-to-end efficiency.** SOMA reduces post-switch cost by serving the adapted surrogate with compressed context, but it also introduces one-time costs from soft-prompt probing and LoRA adaptation. Thus, the key question is whether the total session cost decreases after this overhead is included. SOMA reduces average token usage by avoiding repeated full-history serving after switching (Appendix F.1). Table 4 further shows the amortized operating range: **SOMA is slightly unfavorable for sessions of 1–4 turns (−4.1% latency), breaks even at 5–8 turns (+4.7%), and is clearly beneficial from 9 turns on (+16.3% to +29.1% latency, +23.8% to +37.2% tokens).** This matches SOMA’s intended use case as an amortized serving strategy for conversations where enough later turns remain local to the warm-start state. **Figure 2 reports throughput on all six datasets for both families: SOMA matches or exceeds the speed of Surrogate and History-FT and clearly improves over Original and History-Prefix (dataset-level token costs for every method are reported in Appendix F.1).**

**RQ4: Which components of SOMA are most important?** We first ablate SOMA on the LLaMA family to identify which components drive the gains. Figure 3a shows that full SOMA performs

Table 4: ShareGPT break-even after all overhead.

| Turns | Latency | Tokens |
|-------|---------|--------|
| 1–4   | −4.1%   | −1.2%  |
| 5–8   | +4.7%   | +8.5%  |
| 9–12  | +16.3%  | +23.8% |
| 13+   | +29.1%  | +37.2% |

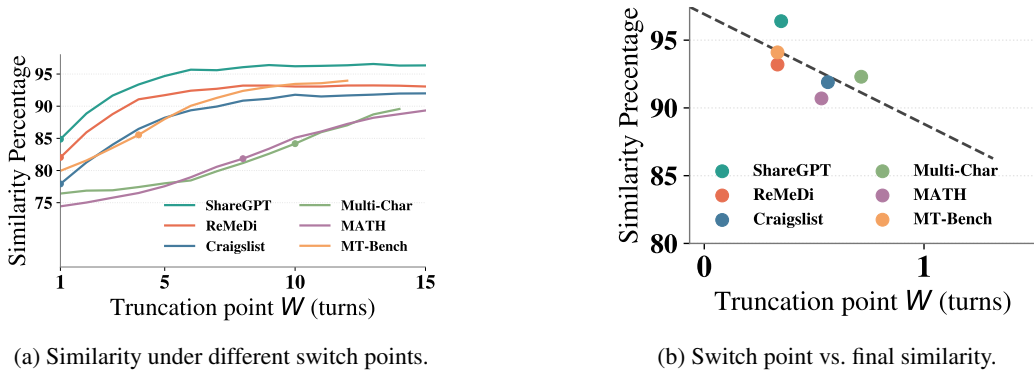


Figure 4: Warm-start and switching behavior. Easier local-dialogue settings switch earlier, while reasoning-heavy or multi-party settings need longer warm starts and achieve lower final similarity.

Table 5: Reliability of session-local serving: later turns stay context-dependent (a), and drift-aware rollback recovers most of the loss under abrupt topic shifts (b).

| (a) MATH context dependency. |                  |                  |                  | (b) Topic-shift stress test. |                 |                 |                                   |        |          |
|------------------------------|------------------|------------------|------------------|------------------------------|-----------------|-----------------|-----------------------------------|--------|----------|
| Setting                      | 1-3              | 4-6              | 7+               | Data                         | Clean           | No RB           | SOMA                              | Detect | False RB |
| Full history                 | 46.92 $\pm$ 0.58 | 47.18 $\pm$ 0.51 | 46.75 $\pm$ 0.49 | ShareGPT                     | 96.4 $\pm$ 1.91 | 88.5 $\pm$ 2.03 | <b>94.1 <math>\pm</math> 1.47</b> | 90.8%  | 4.3%     |
| No history                   | 43.78 $\pm$ 0.66 | 34.92 $\pm$ 0.76 | 24.88 $\pm$ 0.86 | ReMeDi                       | 93.2 $\pm$ 0.98 | 85.7 $\pm$ 1.92 | <b>91.1 <math>\pm</math> 1.34</b> | 89.6%  | 3.9%     |
| Drop                         | 3.14             | 12.26            | 21.87            | Craigslist                   | 91.9 $\pm$ 2.49 | 83.1 $\pm$ 2.11 | <b>89.3 <math>\pm</math> 1.68</b> | 88.9%  | 3.8%     |

best across all datasets. Removing the anti-degeneration loss consistently reduces performance, showing that entropy regularization keeps prompt mining stable and prevents collapsed, low-diversity soft prompts. Removing both the expectation-weighted term and the anti-degeneration loss leads to a larger drop, especially on harder datasets. This result shows that SOMA benefits from both stable prompt search and disagreement-aware mining. Figure 3b repeats the ablation on the Qwen family with the same ordering, so the conclusion does not depend on the backbone. Figure 3c breaks MT-Bench-101 down by ability for the LLaMA family: SOMA improves most on reasoning and questioning, where the original model has the largest advantage over the surrogate, and also improves memory, understanding, rephrasing, and interference handling.

**RQ5: When is session-local serving reliable, and when should SOMA fall back?** We next test when SOMA’s locality assumption holds and how rollback helps when it fails. Table 5a shows that full-history MATH performance stays stable across turn buckets, while no-history performance drops sharply in later turns. This means later turns are not simply easier; they remain strongly dependent on the earlier dialogue context. Table 5b then evaluates abrupt topic shifts. Without rollback, similarity drops substantially. With SOMA’s drift-aware rollback, similarity returns to within 2.1–2.6 points of the clean setting (e.g., 88.5  $\rightarrow$  94.1 vs. 96.4 on ShareGPT), with 88.9–90.8% of shifts detected and only 3.8–4.3% false rollbacks. These results show that SOMA should be viewed as a local serving strategy with automatic fallback. Figure 4 shows the warm-start side: goal-anchored dialogues (ShareGPT, ReMeDi, Craigslist) reach high similarity after a few turns, whereas reasoning-heavy or multi-party ones (MATH, Multi-Char) need longer warm starts and plateau lower, so the switch point must be data-dependent, which is exactly what the acceptance gate implements.

## 6 CONCLUSION

We presented SOMA, an efficient multi-turn LLM serving framework that uses early dialogue turns to adapt a small surrogate to the original model’s session-local behavior. It mines soft-prompt directions, distills them with LoRA, and switches only when alignment is sufficient. Experiments show that SOMA better preserves original-model responses, improves task-grounded quality, and reduces serving cost once warm-start overhead is amortized. These results suggest that session-local surrogate serving is a practical direction for efficient multi-turn LLM deployment when sessions are long and locally coherent enough to amortize the warm-start cost, as Table 4 quantifies. Limitations and future work are discussed in Appendix I.

## AI USE STATEMENT

In this work, we used generative AI tools to improve the readability of the manuscript, such as grammar checking, wording refinement, and formatting suggestions, and to assist with implementing and debugging parts of the experimental code, such as evaluation and plotting scripts. We did not use generative AI tools to propose or refine the research hypotheses, design the methodology or the experiments, develop the theoretical framework, formulate or prove the mathematical claims, generate or clean datasets, or interpret the results. We have reviewed all AI-assisted work: AI-suggested edits were checked by the authors for accuracy and faithfulness to the intended meaning, and AI-assisted code was verified and tested for correctness by the authors before use. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

## ETHICS STATEMENT

This work studies efficient multi-turn LLM serving and does not involve human subjects, user studies, or the collection of new personal data. All experiments use publicly available dialogue and reasoning benchmarks under their respective licenses (Appendix C.1), and we release no new dataset. SOMA is intended to reduce the cost and latency of serving LLMs, which can broaden access to interactive LLM systems and reduce their energy footprint. As with any method that substitutes a smaller surrogate for a larger model, there is a risk of quality degradation when a conversation drifts away from the adapted local region, which is most consequential in high-stakes domains such as medical, legal, or financial assistance. SOMA mitigates this risk through its semantic acceptance gate and drift-aware rollback, but real deployments should retain monitoring, conservative switching thresholds, and human oversight where errors may cause harm. Because SOMA adapts to dialogue history, deployments should also follow data-minimization and retention practices to protect user privacy. Broader impacts and limitations are discussed further in Appendices H and I.

## REPRODUCIBILITY STATEMENT

We have made efforts to ensure that our results are reproducible. An anonymized implementation of SOMA, including scripts for all major experiments, is available at <https://anonymous.4open.science/r/SOMA-4175>. Appendix C.1 describes the datasets and Appendix C.4 the filtering of non-context-dependent dialogues; Appendix C.2 gives the implementation details and hyperparameter ranges of SOMA; Appendix C.3 specifies the baselines; Appendix C.5 describes the LLM-judge protocol; and Appendix C.6 lists the software packages and versions used. For the theoretical results in Section 4, all assumptions are stated explicitly and complete proofs are provided in Appendix E, with the supporting mathematical background in Appendix D.

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## APPENDIX

## A NOTATIONS

This section summarizes all notations used throughout this paper.

Table 6: Notation summary.

| Symbol                                   | Meaning                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------|
| $\mathcal{D}_k$                          | Dialogue prefix $\{(q_t, a_t)\}_{t=1}^k$                                        |
| $q_{\leq t}$                             | Full history up to turn $t$                                                     |
| $F, G$                                   | Original (large) model; surrogate (small) model                                 |
| $\mathbf{h}_t = f_M(q_{\leq t})$         | Hidden state at turn $t$ for model $M$                                          |
| $\mathcal{H}_k$                          | $\{\mathbf{h}_1, \dots, \mathbf{h}_k\}$ , first- $k$ hidden states              |
| $\mathcal{M}_k^M$                        | Local response manifold induced by $\mathcal{D}_k$ under $M$                    |
| $\mathcal{V}, E$                         | $G$ 's vocabulary; embedding matrix $E \in \mathbb{R}^{d \times  \mathcal{V} }$ |
| $\Pi_t$                                  | Next-token distribution of $G$ at turn $t$ ; entry $\Pi_t[v]$                   |
| $\mathbf{e}_v$                           | Embedding of token $v$ (column of $E$ )                                         |
| $\mathbf{P} \in \mathbb{R}^{L \times d}$ | Soft prompt (row-wise prompt tokens in embedding space)                         |
| $\mathbf{V}(\mathbf{P})$                 | Verbalization for prompt $\mathbf{P}$                                           |
| $N_k(u)$                                 | $k$ nearest neighbors of token $u$ by cosine in $E$                             |
| $\bar{\mathbf{e}}_{t,i}$                 | Expected embedding $\bar{\mathbf{e}}_{t,i} = E^\top \Pi_{t,i}$ at position $i$  |
| $w_{t,i}$                                | Expectation-weighted factor for semantic divergence loss                        |
| $ B _{\text{eff}}$                       | Effective batch size (accounts for dependence)                                  |
| $\Delta$                                 | Empirical improvement (old–new) on a post-warm-start batch                      |

## B RELATED WORK

**Efficient LLM for Multi-turn Dialogues.** Recent efforts to improve multi-turn LLM serving efficiency fall into two main paradigms. The first involves **single-model approaches** that reduce context length or reuse computation. These include *summarization and context compression* (Wang et al., 2025; Chen et al., 2024b; Xiao et al., 2024), *memory augmentation* (Melz, 2023; Gutiérrez et al., 2024), and *caching and attention reuse*, such as **CachedAttention** (Gao et al., 2024), **prompt caching** (Anthropic, 2024), **resource-aware multi-turn scheduling** (Jeong & Ahn, 2025), **isolated per-turn KV compression in FlowKV** (Liu et al., 2025), and **adaptive prefill sparsification with progressive KV compression in LoopServe** (Li et al., 2025a). While effective in lowering per-turn cost, these methods still rely on repeated large-model inference, which also leads to high API expenses, latency, and GPU demand. Moreover, they may truncate or underutilize dialogue context, harming performance on complex tasks. The second paradigm adopts **multi-model approach**, using smaller models for simple queries and escalating difficult ones to larger LLMs (Behera et al., 2025; Schick et al., 2023; Ding et al., 2024), typically via model routing (Shnitzer et al., 2023; Ong et al., 2024), **multi-round routing and aggregation learned with reinforcement learning** (Zhang et al., 2025), and distillation (Hinton et al., 2015). However, pre-trained small models often generalize poorly on complex multi-turn dialogues, while switching models adds inefficiency.

**Relation to systems-level serving and test-time adaptation.** Prefix and KV caching (CachedAttention, prompt caching, FlowKV, LoopServe) reduce the prefill cost of re-processing the history on the *same* large model; the per-token decode cost, memory footprint, and API price of that model are unchanged. SOMA is orthogonal: after the switch it removes the large model from the serving path, and the surrogate can itself be served with prefix caching. Our efficiency comparison already serves every method, including Original, through vLLM with automatic prefix caching enabled (Section 5), so the reported token and latency gains are measured on top of KV-cache reuse. Test-time adaptation methods such as Temp-LoRA (Wang et al., 2024) and StreamAdapter (Muhtar et al., 2024) adapt a model to its *own* generated or in-context text; History-FT and SOMA instead perform cross-model distillation, adapting the surrogate to the large model’s warm-start responses within the session.

**Local Manifold Approximation.** We use “local manifold approximation” in the informal sense of classical local methods such as LLE (Roweis & Saul, 2000), LTSA (Zhang et al., 2007), and Laplacian eigenmaps (Belkin & Niyogi, 2003): a dialogue prefix induces a local region of the model’s response behavior, and SOMA approximates the large model only within that region rather than globally. SOMA does not compute intrinsic dimensions, tangent spaces, or geodesics. The local, first-order view enters the method only through the Jacobian and Fisher structure in Theorem 3.2 and the local quadratic assumption in Section 4, and we treat it as a modeling assumption whose usefulness is tested empirically.

## C REPRODUCIBILITY

In this section, we introduce the details of the experiments in this paper for reproducibility. At the same time, we have uploaded all necessary code to our GitHub repository to reproduce the results presented in this paper: <https://anonymous.4open.science/r/SOMA-4175>. All major experiments are encapsulated as shell scripts, which can be conveniently executed. We introduce the details for reproducibility in the subsections below.

### C.1 REAL-WORLD DATASETS

In this section, we briefly present the real-world datasets used in this paper, and all these datasets are commonly used datasets in multi-turn conversation tasks. *ShareGPT* (Chen et al., 2024a) is a large-scale collection of high-quality image–text conversations and captions. *ReMeDi* (Yan et al., 2022) is a multi-domain Chinese medical dialogue corpus of doctor–patient conversations. *Craigslist* (He et al., 2018) contains multi-turn buyer–seller negotiation chats from Craigslist, enabling study of bargaining strategies and goal-directed dialogue. *Multi-Char* (agentlans, 2024) provides multi-character conversational scenarios with role specifications to evaluate coordination and role consistency in multi-party dialogue. *MATH* (Hendrycks et al., 2021) is a benchmark of competition-style math problems with step-by-step solutions designed to assess mathematical reasoning in language models. *MT-Bench* (Zheng et al., 2023) is a multi-turn benchmark to assess response quality across diverse tasks. In this study, we filter out the non-context-dependent dialogues in these datasets.

### C.2 IMPLEMENTATION OF SOMA

We implement SOMA based on PyTorch with HuggingFace Transformers, serve inference via vLLM with FlashAttention (Shah et al., 2024), and run on one node with  $4 \times 80\text{G A100 GPUs}$ . Soft-prompt tuning optimizes only the prompt tensor  $\mathbf{P} \in \mathbb{R}^{L \times d}$  on the surrogate  $G$  using AdamW, cosine decay, gradient clipping, and a KV cache for a single forward pass per turn. The objective combines unlikelihood on a semantic neighborhood, an expectation-weighted penalty using the truncated top- $m$  expectation, and a light anti-degeneration entropy term. The mined prompt–response pairs are then used to adapt  $G$  with LoRA on attention projections (rank  $r$ ), keeping the base weights frozen; early stopping is triggered by validation divergence. At inference, a cosine closeness gate with a lightweight sentence encoder decides a one-time switch to the adapted surrogate; the window  $W$  and acceptance batch  $|B|$  follow the detection bound, the number of parallel candidates  $M$  follows the coverage guarantee, and  $(k, m)$  follow the efficiency analysis. Prompt length  $L \in \{4, 8, 16, 32, 64\}$ ; learning rate  $\eta \in [1 \times 10^{-4}, 5 \times 10^{-3}]$ ; AdamW weight decay  $\lambda_{\text{wd}} \in [0, 10^{-2}]$ ; gradient clip  $\in [0.5, 1.0]$ ; neighborhood size  $k \in \{20, 50, 100\}$ ; expectation truncation  $m \in \{50, 100, 200\}$ ; temperature  $\tau \in [0.4, 1.2]$ ; expectation weight  $\lambda \in [0.5, 2.0]$ ; anti-degeneration weight  $\beta \in [0.02, 0.15]$ ; LoRA rank  $r \in \{4, 8, 16, 32\}$  with scale  $\alpha_{\text{lor}} \in \{4, 8, 16, 32\}$ ; LoRA LR  $\eta_{\text{lor}} \in [5 \times 10^{-5}, 2 \times 10^{-3}]$ ; warm-start window  $W \in [3, 12]$  turns; acceptance batch  $|B| \in [6, 24]$  contexts; parallel candidates  $M \in \{3, 4, 5\}$ ; switch threshold  $\varepsilon \in [0.05, 0.12]$  with  $m_{\text{cons}} \in \{2, 3\}$ .

### C.3 IMPLEMENTATION OF BASELINES

**Original:** We query the large model (family-appropriate: LLaMA-3.1-70B or Qwen-3-8B) with the full dialogue history at every turn to get the responses.

**Surrogate:** We query the small model (LLaMA-2-7B or Qwen-3-0.6B) with the full dialogue history at every turn to get the responses.

You are an expert conversation analyst.

Task – Classification:

\* **context-dependent** \*: The response of every turn relies on information from earlier turns (the conversation is a single chain of related reasoning steps).

\* **context-independent** \*: The response of every turn is independent of earlier turns (the conversation is a collection of independent questions and answers).

Conversation (verbatim):

{dialogue}

Question: Is the conversation context-dependent or context-independent?

Please answer with exactly one word:

Either **context-dependent** or **context-independent** — nothing else.

Figure 5: Instructions for the LLM to filter context-dependent dialogue.

**History-Prefix:** The surrogate receives the entire conversation produced by the Original up to  $t-1$  and generates  $a_t$ ; no parameter updates are performed. The switching is the same as SOMA.

**History-FT:** We fine-tune the surrogate on  $(S_t, a_t^F)$  pairs where  $S_t$  is the Original’s full context up to  $t$  and  $a_t^F$  is the next reply, using LoRA on attention projections with early stopping; inference then runs the fine-tuned surrogate without the Original. The switching is the same as SOMA.

**LLMLingua-2** (Pan et al., 2024): We compress the Original’s history at each turn with LLMLingua-2 and feed the compressed summary to the surrogate; we follow the authors’ recommended settings.

**RouteLLM** (Ong et al., 2024): We adopt the released router to choose between the Original and the surrogate per turn (complex vs. simple queries), following the original settings from the paper.

#### C.4 DATA FILTERING

To ensure that SOMA is evaluated on context-dependent dialogues, we prompt multiple strong LLMs, including GPT-OSS, DeepSeek-V3, and Gemma-2-27B, with the classification prompt shown in Figure 5, and retain only conversations that all three models recognize as context-dependent. This avoids model-specific biases and yields a high-quality subset of dialogues where later turns meaningfully depend on earlier turns, matching the use cases in this paper.

#### C.5 IMPLEMENTATION OF LLM JUDGE

The prompt for the LLM judge is shown in Figure 6. It adapts the single-answer grading protocol of MT-Bench (Zheng et al., 2023) and MT-Bench-101 (Bai et al., 2024) to reference-based similarity scoring: the judge receives the original model’s response as the reference and scores the surrogate’s response on a 0–1 scale along accuracy, reasoning, context integration, and communication. The reference-based framing and the 0–1 scale are our modifications; the MT-Bench protocols grade single answers on a 1–10 scale without a reference. Each response is scored by three judges (GPT-OSS, DeepSeek-V3, and Gemma-2-27B), and we report the mean.

You are an impartial evaluator.  
 You are evaluating the AI assistant’s capabilities in solving a problem posed by ‘Human’.  
 Assess the assistant’s performance based on the following criteria:

1. Accuracy: Verify the correctness of the AI assistant’s answer against the ground truth (reference solution).
2. Reasoning: Assess the completeness, clarity, and logical soundness of the step-by-step reasoning process.
3. Context Integration: Consider whether the assistant integrates relevant prior dialogue or context that influences the solution.
4. Communication: Appraise how clearly and effectively the assistant communicates its reasoning to aid understanding.

Score Guidelines (0.0 to 1.0):

- 1.0: Completely correct answer and meticulously clear, step-by-step reasoning that is logically sound and instructional.
- 0.7–0.9: Correct answer with a well-articulated reasoning process that includes the necessary steps and promotes understanding.
- 0.4–0.6: Partially correct answer with some minor reasoning flaws or omissions.
- 0.0–0.3: Incorrect answer and/or poor reasoning that lacks clarity or logic.

Only respond with a single score between 0.0 and 1.0. Do not include any explanation.  
 ---

Reference Solution:  
 {Original Response}  
 Candidate Answer:  
 {Surrogate Response}

Score:

Figure 6: Instructions for the LLM judge to evaluate the response similarity.

## C.6 PACKAGES REQUIRED FOR IMPLEMENTATION

We perform all experiments on a server equipped with Nvidia A6000 GPUs. Below we list the key packages and their versions used in our implementation:

- **Python** == 3.10
- **pytorch** == 2.8.0 + CUDA 12.8
- **torchvision** == 0.19.0
- **torchaudio** == 2.8.0
- **numpy** == 1.26.x
- **pandas** == 2.2.x
- **scipy** == 1.12.x
- **cmake** == 3.28+
- **ninja** == 1.11+
- **ipython** == 8.x
- **psutil** == 5.9+
- **vllm** == 0.10.2
- **transformers** == 4.56.1
- **accelerate** == 1.9.0
- **bitsandbytes** == 0.46.1
- **sentencepiece** == 0.2.0
- **tiktoken** == 0.11.0
- **einops** == 0.8.1

- **datasets** == 4.0.0
- **huggingface-hub** == 0.34.2
- **safetensors** == 0.5.3
- **ray** == 2.49.1
- **scikit-learn** == 1.7.1
- **fastapi** == 0.116.2
- **uvicorn** == 0.35.0

## D MATHEMATICAL CONCEPTS

Here we provide a more comprehensive view of the relevant concepts that can be helpful to understand the idea of stratification as discussed in the main body of the paper.

**Definition D.1** (Preimage). Let  $f : X \mapsto Y$  be a function from a set  $X$  (*domain*) to a set  $Y$  (*codomain*). For any subset  $N \subseteq Y$ , the *preimage* of  $N$  under  $f$ , denoted  $f^{-1}(N)$ , is defined as:

$$f^{-1}(N) = \{x \in X \mid f(x) \in N\}.$$

In other words,  $f^{-1}(N)$  consists of all elements in the domain  $X$  that are mapped into the subset  $N$  of the codomain  $Y$ .

**Definition D.2** (Metric Space). A set  $X$ , whose elements are called points, is said to be a *metric space* if for any two points  $p, q \in X$ , there is an associated real number  $d(p, q)$ , called the *distance* from  $p$  to  $q$ , such that:

1.  $d(p, q) \geq 0$ , and  $d(p, q) = 0 \iff p = q$ ;
2.  $d(p, q) = d(q, p)$  (symmetry);
3.  $d(p, q) \leq d(p, r) + d(r, q)$  for any  $r \in X$  (triangle inequality).

Any function satisfying these properties is called a *distance function*, or a *metric*.

**Definition D.3** (Neighborhood). Let  $X$  be a metric space. A set  $N_r(p) \subset X$  is called a *neighborhood* of a point  $p \in X$  if it consists of all points  $q \in X$  such that  $d(p, q) < r$  for some radius  $r > 0$ . The number  $r$  is called the *radius* of the neighborhood.

**Definition D.4** (Continuous). Let  $X$  and  $Y$  be two topological spaces. A function  $f : X \mapsto Y$  is *continuous* if for each point  $x \in X$  and each neighborhood  $N$  of  $f(x)$  in  $Y$ , the set  $f^{-1}(N)$  is a neighborhood of  $x \in X$ .

**Definition D.5** (Topological Equivalence or Homeomorphism). A function  $h : X \mapsto Y$  is called a *homeomorphism* if it is one-to-one, continuous, and has a continuous inverse function. When such a function exists,  $X$  and  $Y$  are called *homeomorphic* (or *topologically equivalent*) spaces.

**Definition D.6** (Open Set). A subset  $U \subseteq X$  is called an *open set* if for every point  $p \in U$ , there exists a neighborhood  $N_r(p) \subseteq U$ . That is, each point in  $U$  has some "wiggle room" around it that still lies entirely within  $U$ .

**Definition D.7** (Countable Base (Second Countability)). Let  $X$  be a topological space. A collection  $\mathcal{B}$  of open subsets of  $X$  is called a *base* (or *basis*) for the topology on  $X$  if for every open set  $U \subseteq X$  and every point  $x \in U$ , there exists a set  $B \in \mathcal{B}$  such that

$$x \in B \subseteq U.$$

If there exists a base  $\mathcal{B}$  that is *countable*, then the space  $X$  is said to be *second countable* or to have a *countable base*.

**Definition D.8** (Hausdorff Space). A topological space with the property that two distinct points can always be surrounded by disjoint open sets is called a *Hausdorff space*.

Essentially, Hausdorff spaces are the spaces where any two points being "far off" is defined.

**Definition D.9** (Manifold). A *manifold* of dimension  $n$  is a second-countable Hausdorff topological space in which each point has a neighborhood homeomorphic to Euclidean space  $\mathbb{R}^n$ .

**Definition D.10** (Smooth Manifold). A *smooth manifold* is a manifold  $\mathcal{M}$  equipped with a collection of coordinate charts (i.e., homeomorphisms  $\varphi : U \rightarrow \mathbb{R}^n$  for open sets  $U \subseteq \mathcal{M}$ ) such that all *transition maps* between overlapping charts,

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j),$$

are infinitely differentiable (i.e.,  $C^\infty$ ). This structure is known as a *smooth atlas*, and it allows calculus to be performed on the manifold.

## E THEORETICAL RESULTS AND PROOFS

We first model weak dependence in the dialogue stream via an *Effective Sample Size*. For a bounded, stationary sequence  $\{Z_t\}$  with lag- $\ell$  autocorrelation  $\rho(\ell)$ , define

$$N_{\text{eff}} := \frac{N}{1 + 2 \sum_{\ell=1}^{N-1} \left(1 - \frac{\ell}{N}\right) \rho(\ell)} \in (0, N].$$

when  $\rho(\ell) \equiv 0$ ,  $N_{\text{eff}} = N$ ; positive correlation reduces  $N_{\text{eff}}$ .

### Directional Recovery for the Expectation-weighted Loss

**Statement: Theorem 3.2.** Under local smoothness, bounded expectation weights in Eq. (2), and a rank- $r$  discrepancy for the discrepancy Fisher  $\mathbf{C} = \mathbb{E}[\mathbf{J}(\mathbf{P})^\top \mathbf{J}(\mathbf{P})]$ , any minimizer  $\hat{\mathbf{P}}$  of Eq. (2) has row span that captures at least a  $(1 - \varepsilon)$  fraction of the top- $r$  eigenmass of  $\mathbf{C}$ , with  $\varepsilon \rightarrow 0$  as the window size growing and neighborhood size  $k$  increasing within the local region.

**Proof of Theorem 3.2. Step 1: Per-Event Loss and Local Expansion.** Index events by  $\mathbf{z} = (t, i)$  (turn and position). Let the per-event loss be

$$\ell(\mathbf{P}; \mathbf{z}) = w(\mathbf{z}) \sum_{v \in \mathcal{S}(\mathbf{z})} s_\tau(v | \mathbf{z}) \left[ -\log(1 - \pi_G(v | \mathbf{z}; \mathbf{P})) \right], \quad \mathcal{S}(\mathbf{z}) = \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F).$$

Write  $p_v(\mathbf{P}; \mathbf{z}) = \pi_G(v | \mathbf{z}; \mathbf{P})$  and expand around  $\mathbf{P} = \mathbf{0}$ . Using the chain rule for the softmax parameterization,

$$\log(1 - p_v) = \log(1 - p_v|_0) - \frac{1}{1 - p_v|_0} (\nabla p_v)^\top \text{vec}(\mathbf{P}) - \frac{1}{2} \frac{p_v|_0}{(1 - p_v|_0)^2} (\text{vec}(\mathbf{P})^\top \nabla \log p_v)^2 + o(\|\mathbf{P}\|^2),$$

where  $\nabla$  denotes the gradient w.r.t.  $\text{vec}(\mathbf{P})$  and we used the softmax identity  $\nabla p_v = p_v \nabla \log p_v$ . Summing over  $v \in \mathcal{S}(\mathbf{z})$  with bounded weights  $w(\mathbf{z}) s_\tau(\cdot | \mathbf{z})$  cancels the *linear* term due to local stationarity under aligned-prefix conditioning (the gradient at  $\mathbf{0}$  integrates to zero across the semantic neighborhood; this is the standard property behind Gauss–Newton/Fisher approximations). Thus the second-order term dominates:

$$\ell(\mathbf{P}; \mathbf{z}) = \frac{1}{2} \|\mathbf{J}_z \text{vec}(\mathbf{P})\|_2^2 + o(\|\mathbf{P}\|^2),$$

where  $\mathbf{J}_z$  stacks rows  $\sqrt{w(\mathbf{z}) s_\tau(v | \mathbf{z})} \nabla \log p_v(\mathbf{0}; \mathbf{z})^\top$  for  $v \in \mathcal{S}(\mathbf{z})$ .

**Step 2: Summation over the Window and Fisher Form.** Sum over  $\mathbf{z}$  in the initial window and take expectation (over the empirical distribution of aligned-prefix contexts). We obtain the Gauss–Newton surrogate

$$\mathcal{L}(\mathbf{P}) = \frac{1}{2} \text{vec}(\mathbf{P})^\top \mathbf{C} \text{vec}(\mathbf{P}) + o(\|\mathbf{P}\|^2), \quad \mathbf{C} = \mathbb{E}[\mathbf{J}(\mathbf{0})^\top \mathbf{J}(\mathbf{0})],$$

which is a discrepancy Fisher matrix with importance weights folded into  $\mathbf{J}$ .

**Step 3: Row-Span Parametrization and KY Fan.** Let  $\mathbf{P} \in \mathbb{R}^{L \times d}$ . Any  $\mathbf{P}$  factorizes as  $\mathbf{P} = \mathbf{A}\mathbf{U}^\top$  with  $\mathbf{U} \in \mathbb{R}^{d \times L}$  having orthonormal columns spanning the row space and  $\mathbf{A} \in \mathbb{R}^{L \times L}$ . Then

$$\text{vec}(\mathbf{P}) = (\mathbf{U} \otimes \mathbf{I}_L) \text{vec}(\mathbf{A}), \quad \mathcal{L}(\mathbf{P}) = \frac{1}{2} \text{vec}(\mathbf{A})^\top (\mathbf{U}^\top \mathbf{C} \mathbf{U} \otimes \mathbf{I}_L) \text{vec}(\mathbf{A}) + o(\|\mathbf{A}\|^2).$$

Including the ridge  $\lambda \|\mathbf{P}\|_F^2 = \lambda \|\mathbf{A}\|_F^2$  from the main loss, the minimum over  $\mathbf{A}$  for fixed  $\mathbf{U}$  is proportional to  $\text{Tr}(\mathbf{U}^\top \mathbf{C} \mathbf{U})$ . Maximizing  $\text{Tr}(\mathbf{U}^\top \mathbf{C} \mathbf{U})$  over  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_L$  is solved by the top- $L$  eigenvectors of  $\mathbf{C}$  (Ky Fan’s variational principle). Hence the *optimal row span* equals the top- $L$  eigenspace of  $\mathbf{C}$ .

**Step 4: Empirical Approximation and Eigenspace Stability.** We work with an empirical Fisher  $\hat{\mathbf{C}}$  formed from finitely many events and finite  $k$ . Under bounded weights and local smoothness,  $\|\hat{\mathbf{C}} - \mathbf{C}\|_{\text{op}} \rightarrow 0$  as the window size grows; increasing  $k$  within the local isotropy region reduces variance and retains locality. Davis–Kahan perturbation then gives that the top- $r$  eigenspaces of  $\hat{\mathbf{C}}$  and  $\mathbf{C}$  are close, with principal angles bounded by  $O(\|\hat{\mathbf{C}} - \mathbf{C}\|_{\text{op}}/\text{gap})$ , where  $\text{gap}$  is the spectral gap below  $\lambda_r(\mathbf{C})$ . Consequently, the row span of any empirical minimizer  $\hat{\mathbf{P}}$  captures at least a  $(1 - \varepsilon)$  fraction of the top- $r$  eigenmass of  $\mathbf{C}$  with  $\varepsilon \rightarrow 0$  as the window grows and  $k$  increases up to the local isotropy scale.  $\square$

#### When to Switch: Warm-Start Generalization and Batch Detection.

**Lemma E.1** (Warm-Start Generalization). *Let  $h(S) \in [0, 1]$  be any bounded local statistic, such as  $\text{Gap}_\Theta(S)$  or a query-centroid similarity score. Suppose the warm-start stream has effective sample size  $W_{\text{eff}}$ , accounting for temporal dependence. Then with probability at least  $1 - \delta$ ,*

$$\left| \frac{1}{W} \sum_{t=1}^W h(S_t) - \mathbb{E}_{S \sim \mathcal{Q}}[h(S)] \right| \leq \sqrt{\frac{2 \log(2/\delta)}{W_{\text{eff}}}}.$$

Below we write  $\hat{F}_W = \frac{1}{W} \sum_{t=1}^W \text{Gap}(S_t)$  and  $F^* = \mathbb{E}_{S \sim \mathcal{Q}}[\text{Gap}(S)]$  for the case  $h = \text{Gap}$ .

**Proof of Lemma E.1. Step 1 (Centering and boundedness).** Let  $Z_t = \text{Gap}(S_t) - F^* \in [-1, 1]$  with  $\mathbb{E}[Z_t] = 0$ . Then  $\hat{F}_W - F^* = \frac{1}{W} \sum_{t=1}^W Z_t$ .

**Step 2: Variance Proxy under Dependence.** For stationary  $\{Z_t\}$  with autocovariance  $\gamma(\ell) = \text{Cov}(Z_t, Z_{t+\ell})$ , we have

$$\text{Var}\left(\frac{1}{W} \sum_{t=1}^W Z_t\right) = \frac{1}{W^2} \sum_{t,s=1}^W \gamma(|t-s|) = \frac{1}{W} \left( \gamma(0) + 2 \sum_{\ell=1}^{W-1} \left(1 - \frac{\ell}{W}\right) \gamma(\ell) \right).$$

Let  $\sigma^2 := \gamma(0) \leq 1/4$  (since  $Z_t \in [-1, 1]$ ) and  $\rho(\ell) := \gamma(\ell)/\gamma(0)$  when  $\gamma(0) > 0$ . Then

$$\text{Var}\left(\frac{1}{W} \sum_{t=1}^W Z_t\right) \leq \frac{\sigma^2}{W} \left( 1 + 2 \sum_{\ell=1}^{W-1} \left(1 - \frac{\ell}{W}\right) \rho(\ell) \right).$$

Define the effective size  $W_{\text{eff}}$  as in the preliminaries. Then  $\text{Var}\left(\frac{1}{W} \sum_{t=1}^W Z_t\right) \leq \sigma^2/W_{\text{eff}}$ .

**Step 3: Concentration with Effective Size.** A Hoeffding/Rio inequality for bounded, weakly dependent sequences yields

$$\Pr\left(|\hat{F}_W - F^*| \geq u\right) \leq 2 \exp\left(-\frac{2u^2 W_{\text{eff}}}{(b-a)^2}\right), \quad a = -1, b = 1.$$

Thus  $\Pr(|\hat{F}_W - F^*| \geq u) \leq 2 \exp(-2u^2 W_{\text{eff}})$ . Set the right-hand side to  $\delta$  and solve for  $u$  to obtain  $u = \sqrt{(2 \log(2/\delta))/W_{\text{eff}}}$ .  $\square$

**Statement: Theorem 4.1.** Let  $\Delta = \hat{F}_B^{\text{old}} - \hat{F}_B^{\text{new}}$  on a batch  $B$  of effective size  $|B|_{\text{eff}}$ . Assume  $\Delta - \mathbb{E}[\Delta]$  is sub-exponential with proxy  $(\nu, b)$ :  $\mathbb{E}[\exp(\lambda(\Delta - \mathbb{E}[\Delta]))] \leq \exp(\frac{\lambda^2 \nu^2}{2})$  for  $|\lambda| \leq 1/b$ . Then for any  $\varepsilon > 0$  and  $\delta \in (0, 1)$ ,

$$|B|_{\text{eff}} \geq \frac{2\nu^2}{\varepsilon^2} \log \frac{1}{\delta} + \frac{2b}{3\varepsilon} \log \frac{1}{\delta} \implies \Pr(\Delta \geq \varepsilon) \geq 1 - \delta.$$

**Proof of Theorem 4.1. Step 1 (Bernstein tail bound).** For sub-exponential  $X := \Delta - \mathbb{E}[\Delta]$  with proxy  $(\nu, b)$ ,

$$\Pr(X \leq -t) \leq \exp\left(-\frac{t^2}{2\nu^2 + 2bt/3}\right), \quad t > 0.$$

This is the standard one-sided Bernstein inequality derived from the MGF condition.

**Step 2: From Mean Gain to High-Probability Gain.** We seek  $\Pr(\Delta \geq \varepsilon) \geq 1 - \delta$ . Write

$$\Pr(\Delta < \varepsilon) = \Pr(\Delta - \mathbb{E}[\Delta] < \varepsilon - \mathbb{E}[\Delta]) = \Pr(X < -t), \quad t := \mathbb{E}[\Delta] - \varepsilon.$$

If the *expected* gain satisfies  $\mathbb{E}[\Delta] \geq \varepsilon$  then  $t \geq 0$  and

$$\Pr(\Delta < \varepsilon) \leq \exp\left(-\frac{t^2}{2\nu^2 + 2bt/3}\right) \leq \exp\left(-\frac{\varepsilon^2}{2\nu^2 + 2b\varepsilon/3}\right),$$

using monotonicity of  $t \mapsto t^2/(2\nu^2 + 2bt/3)$  on  $t \geq 0$ . To make this  $\leq \delta$  it suffices that

$$\frac{\varepsilon^2}{2\nu^2 + 2b\varepsilon/3} \geq \log(1/\delta) \iff 2\nu^2 \log(1/\delta) + \frac{2b}{3}\varepsilon \log(1/\delta) \leq \varepsilon^2.$$

Rearranging gives the stated sufficient condition on  $|B|_{\text{eff}}$  once we note that the proxies  $\nu^2, b$  scale as  $1/|B|_{\text{eff}}$  for averages. Equivalently, write  $\nu^2 = \tilde{\nu}^2/|B|_{\text{eff}}$  and  $b = \tilde{b}/|B|_{\text{eff}}$  for single-sample proxies  $(\tilde{\nu}^2, \tilde{b})$ , then solve for  $|B|_{\text{eff}}$ :

$$|B|_{\text{eff}} \geq \frac{2\tilde{\nu}^2}{\varepsilon^2} \log \frac{1}{\delta} + \frac{2\tilde{b}}{3\varepsilon} \log \frac{1}{\delta}.$$

We re-denote  $(\tilde{\nu}^2, \tilde{b})$  as  $(\nu^2, b)$  in the theorem statement.  $\square$

**Corollary E.2 (Switching Rule).** Choose  $W$  so that the warm-start uncertainty in Lemma E.1 is at most  $\eta$  (i.e.,  $W_{\text{eff}} \geq 2 \log(2/\delta)/\eta^2$ ), and choose  $|B|_{\text{eff}}$  so that  $\gamma_B(\delta) \leq \gamma$  in Theorem 4.1. Switching is justified when the adapted surrogate passes both the output-fidelity gate and the locality gate with margins larger than  $\eta + \gamma$ . The combined decision error is controlled at confidence at least  $1 - 2\delta$  by a union bound.

**Proof of Corollary E.2.** Lemma E.1 ensures  $|\hat{F}_W - F^*| \leq \eta$  with prob.  $\geq 1 - \delta$  when  $W_{\text{eff}} \geq 2 \log(2/\delta)/\eta^2$ . Theorem 4.1 ensures  $\Pr(\Delta \geq \varepsilon) \geq 1 - \delta$  when the batch bound holds. By a union bound, the probability that both events hold is at least  $1 - 2\delta$ . If both hold, the total decision error (warm-start approximation plus detection slack) is at most  $\eta + \varepsilon$ .  $\square$

## Coverage and Suboptimality to Determine the Number of Soft-Prompt Candidates.

**Statement: Theorem 4.2.** Let the active local subspace be  $r_{\text{act}}$ -dimensional with  $r_{\text{act}} \geq 2$ , and draw  $M$  i.i.d. unit vectors  $\mathbf{u}_1, \dots, \mathbf{u}_M \sim \text{Unif}(\mathbb{S}^{r_{\text{act}}-1})$ . Let  $p_{r_{\text{act}}}(\theta) = \frac{1}{2} I_{\sin^2 \theta}(\frac{r_{\text{act}}-1}{2}, \frac{1}{2})$  be the spherical-cap probability of falling within angle  $\theta \in (0, \pi/2]$  of the dominant direction  $\mathbf{v}_1$ , where  $I_x(\cdot, \cdot)$  is the regularized incomplete beta function. Then

$$\Pr\left(\min_{m \leq M} \angle(\mathbf{u}_m, \mathbf{v}_1) \leq \theta\right) = 1 - (1 - p_{r_{\text{act}}}(\theta))^M \geq 1 - \left(1 - (\sin \theta)^{r_{\text{act}}-1}\right)^M.$$

For  $r_{\text{act}} = 1$ , one candidate suffices.

**Proof of Theorem 4.2. Step 1: Exact cap probability.** WLOG set  $\mathbf{v}_1$  as the north pole. For  $\mathbf{U} \sim \text{Unif}(\mathbb{S}^{r_{\text{act}}-1})$ , the random variable  $T = \langle \mathbf{U}, \mathbf{v}_1 \rangle$  has density  $f_T(t) \propto (1 - t^2)^{(r_{\text{act}}-3)/2}$  on  $t \in [-1, 1]$ . Hence  $\Pr(\angle(\mathbf{U}, \mathbf{v}_1) \leq \theta) = \Pr(T \geq \cos \theta) = I_{\sin^2 \theta}(\frac{r_{\text{act}}-1}{2}, \frac{1}{2})$ .

**Step 2: Lower Bound** For  $\theta \in (0, \pi/2]$ ,  $\sin \theta \in (0, 1]$  and we have the elementary bound

$$I_{\sin^2 \theta}(\frac{r_{\text{act}}-1}{2}, \frac{1}{2}) \geq (\sin \theta)^{r_{\text{act}}-1}.$$

(Proof:  $I_x(a, b) = \frac{1}{B(a, b)} \int_0^x t^{a-1} (1-t)^{b-1} dt \geq \frac{1}{B(a, b)} \int_0^x t^{a-1} dt = \frac{x^a}{a B(a, b)} \geq x^a$  since  $a B(a, b) \leq 1$  for  $a \geq 1/2, b \geq 1/2$ ; let  $x = \sin^2 \theta$  and  $a = (r_{\text{act}} - 1)/2$ .)

**Step 3: Independence across  $M$  Samples.** Thus, for one sample,  $p_\theta := \Pr(\angle(\mathbf{U}, \mathbf{v}_1) \leq \theta) \geq (\sin \theta)^{r_{\text{act}}-1}$ . The probability *none* among  $M$  falls in the cap is  $(1 - p_\theta)^M \leq (1 - (\sin \theta)^{r_{\text{act}}-1})^M$ . Therefore

$$\Pr\left(\min_{m \leq M} \angle(\mathbf{u}_m, \mathbf{v}_1) \leq \theta\right) = 1 - (1 - p_\theta)^M \geq 1 - \left(1 - (\sin \theta)^{r_{\text{act}}-1}\right)^M.$$

□

**Lemma E.3** (Directional Suboptimality). *Let  $\mathbf{H}_T \succeq 0$  have largest eigenvalue  $\lambda_1$  and top eigenvector  $\mathbf{v}_1$ . For any unit  $\hat{\mathbf{u}}$  with  $\angle(\hat{\mathbf{u}}, \mathbf{v}_1) \leq \theta$ ,*

$$\hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} \geq \lambda_1 \cos^2 \theta, \quad \lambda_1 - \hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} \leq \lambda_1 \sin^2 \theta.$$

**Proof of Lemma E.3.** Decompose  $\hat{\mathbf{u}} = \cos \theta \mathbf{v}_1 + \sin \theta \mathbf{w}$  with  $\|\mathbf{w}\|_2 = 1$  and  $\mathbf{w} \perp \mathbf{v}_1$ . Then

$$\hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} = \lambda_1 \cos^2 \theta + \sum_{j \geq 2} \lambda_j \langle \mathbf{v}_j, \mathbf{w} \rangle^2 \sin^2 \theta \geq \lambda_1 \cos^2 \theta,$$

since  $\lambda_j \geq 0$ . Subtract from  $\lambda_1$  to obtain the residual bound.

□

**Corollary E.4** (Candidate Budget). *For  $r_{\text{act}} \geq 2$ , to obtain coverage probability at least  $1 - \delta$  within angle  $\theta$  in Theorem 4.2, it suffices to choose  $M \geq \lceil \log \delta / \log(1 - p_{r_{\text{act}}}(\theta)) \rceil$ , where  $p_{r_{\text{act}}}(\theta) = \frac{1}{2} I_{\sin^2 \theta}(\frac{r_{\text{act}}-1}{2}, \frac{1}{2})$  and  $I_x(\cdot, \cdot)$  is the regularized incomplete beta function. Together with Lemma E.3, this gives an explicit tradeoff between candidate count and local directional quality.*

**Proof of Corollary E.4.** We require  $1 - (1 - (\sin \theta)^{r_{\text{act}}-1})^M \geq 1 - \delta$ . Equivalently  $(1 - (\sin \theta)^{r_{\text{act}}-1})^M \leq \delta$ , giving

$$M \geq \frac{\log(1/\delta)}{\log((1 - (\sin \theta)^{r_{\text{act}}-1})^{-1})}.$$

□

## F ADDITIONAL EXPERIMENTAL RESULTS

This appendix provides supporting results that complement the main experiments: dataset-level token costs, the effect of warm-start length on switching, and the empirical variance concentration behind the switching bound.

### F.1 ADDITIONAL EFFICIENCY RESULTS

The main text reports the amortized break-even analysis and throughput. Here we provide the dataset-level token results. Figure 7 shows SOMA’s average tokens per dialogue, and Figure 8 shows that SOMA consistently uses fewer tokens per dialogue across both model families. The reason is that after switching, SOMA serves later turns with the adapted surrogate and a compressed context rather than repeatedly sending the full growing history. This avoids the large token cost of Original and History-Prefix while retaining stronger response similarity than directly using the surrogate.

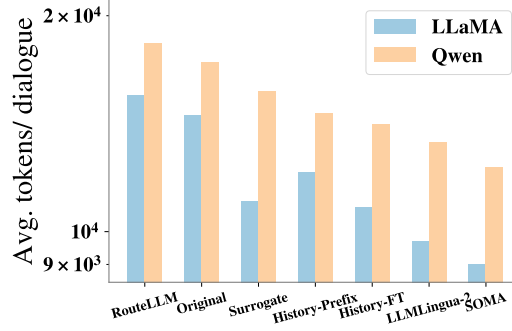


Figure 7: Average tokens per dialogue for SOMA on both model families across the six datasets.

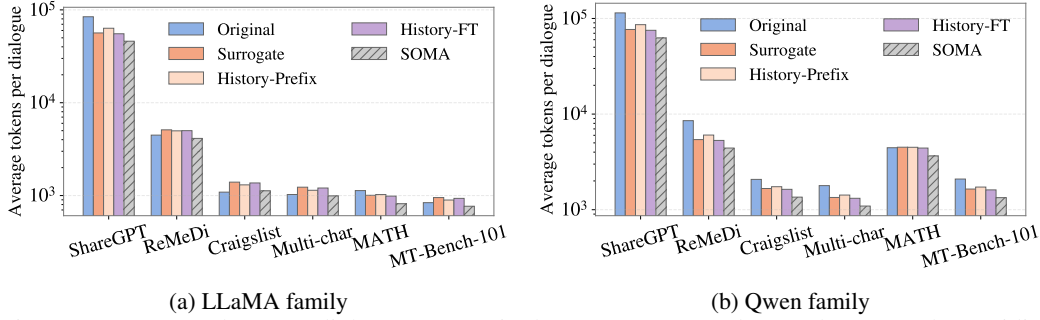


Figure 8: Average tokens per dialogue across six datasets. SOMA reduces token usage by avoiding repeated full-history serving after switching.

## F.2 WARM-START LENGTH AND SWITCHING BEHAVIOR

The main text (Figure 4) reports how final similarity depends on the switch point. The sweep uses warm-start windows  $W \in [1, 15]$  and measures the final response similarity after switching. Once the curve flattens, adding more warm-start turns brings little quality gain but delays the efficiency benefit, so short warm starts suffice for general or goal-anchored chats, while compositional reasoning and multi-party dialogues need more evidence before switching.

## F.3 EMPIRICAL VARIANCE CONCENTRATION

Finally, we examine how the variance of surrogate–teacher similarity changes with the warm-start window  $W$ . For each dataset, we compute the empirical standard deviation across dialogues at each truncation point. Figure 9 shows that variance decreases as  $W$  grows: early turns are highly variable, while larger warm-start windows enter a more stable regime. This trend is consistent with the concentration behavior used in Lemma E.1 and the switching bound in Theorem 4.1. In other words, early turns provide the largest information gain, while later warm-start turns mainly reduce uncertainty until the switching decision becomes stable.

## G DESIGN RATIONALE AND SERVING CONSIDERATIONS

This section collects design questions raised during review that the main text could only touch on briefly.

**Why soft prompts, and why adversarial?** Soft prompts are a convenient probe for three practical reasons: they are differentiable, so a handful of gradient steps on the surrogate  $G$  can search a continuous neighborhood of the warm-start context instead of enumerating discrete paraphrases; they live in  $G$ ’s own embedding space, so the semantic neighborhood  $\mathcal{N}_k(\cdot)$  and the expected-embedding weight are computed with the same token embeddings and reuse the logits of a single forward pass; and the ANN-based neighborhood keeps the per-step cost proportional to  $k + m$  rather than

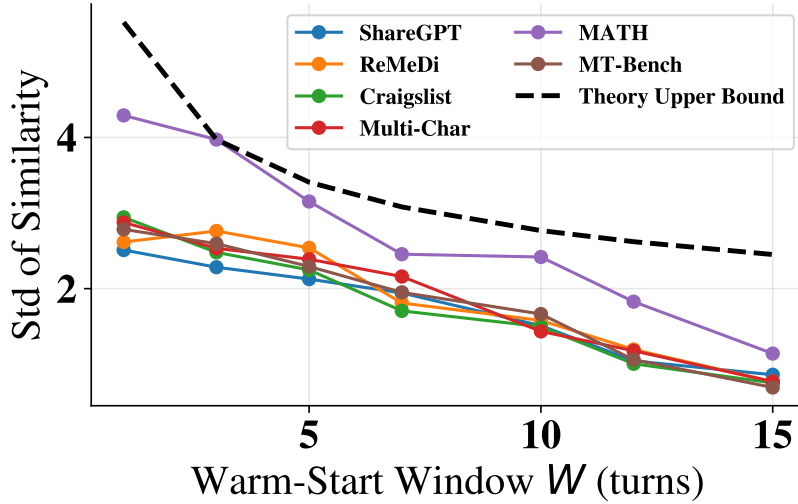


Figure 9: Variance of response similarity versus warm-start window  $W$ . Larger warm-start windows reduce variance and support more stable switching decisions.

$|\mathcal{V}|$ . Alternative perturbations (input noise, paraphrasing, or adversarial token substitution) can also explore beyond the finite warm-start turns, but they either require an external paraphraser or a discrete search, and they do not directly expose which *directions* in  $G$ 's input space make  $G$  diverge from  $F$ . We maximize rather than minimize divergence because the goal of the mining stage is to find hard cases, not to align  $G$ . A constructive (divergence-minimizing) prompt would fit the few warm-start turns and yield a signal that is confounded with the limited capacity of the prompt itself; an adversarial prompt instead ranks warm-start turns by how easily  $G$  can be pushed away from  $F$  around them, and Theorem 3.2 shows that this ranking is aligned with the dominant directions of the local discrepancy Fisher matrix. The mined prompts are discarded after adaptation; only the hardness scores  $r_t$  enter Eq. 4.

**Relation to divergence-weighted History-FT and Random-FT.** The LoRA objective in Eq. 4 already weights every warm-start turn by its hardness  $\omega_t \propto r_t$ , so SOMA can be read as a divergence-weighted History-FT in which the weights come from prompt mining rather than from the raw output divergence between  $G$  and  $F$  on the warm-start turn. The two weightings differ in what they measure: raw output divergence scores a turn by how far the *unperturbed* surrogate already is from the teacher, whereas the mined score measures how *unstable* the surrogate is around that turn under small input perturbations, which is the quantity Theorem 3.2 relates to the local discrepancy geometry. The ablation in Figure 3a isolates the contributions of the mining objective's two terms, but it does not include a baseline that weights History-FT by raw output divergence or one that samples training turns uniformly at random with the same LoRA budget; we consider both important controls and leave them to future work.

**Sensitivity, divergence, and the perturbation space.** Theorem 3.2 characterizes what the mining objective recovers; it does not by itself prove that sensitivity to prompt perturbations is a better predictor of adaptation utility than direct output divergence, nor that the continuous perturbation space of  $\mathbf{P}$  covers the space of natural-language future turns. We therefore treat these as modeling assumptions and test them indirectly: the topic-shift stress test (Table 5b) probes future turns that leave the mined region, and the warm-start sweep (Appendix F.2) shows how much of the local behavior is captured as the window grows. A direct empirical comparison of the two weightings is the most informative next step.

**Number of large-model calls and target consistency.** The original model  $F$  is queried once per warm-start turn to serve the user; these  $W$  responses are stored and reused as the mining targets  $a_t^F$ , as the LoRA targets, and as the references for the acceptance gate.  $F$  is not re-queried while  $\mathbf{P}$  is optimized, so the targets are never stale with respect to the dialogue state, and the verbalized prompt  $V(\mathbf{P})$  only affects how the surrogate side of the objective is conditioned. Prompt mining and LoRA

adaptation run entirely on the surrogate and consist of a small number of forward-backward passes over the  $W$  warm-start turns ( $M$  candidate prompts times the mining iterations, followed by the LoRA steps). Their wall-clock cost is part of the end-to-end latency reported in Table 4.

**Drift detection.** The drift statistic is the cosine distance between recent queries and the warm-start centroid, monitored over a sliding window; rollback triggers after  $m_{\text{cons}}$  consecutive violations. Because the statistic is continuous, gradual drift lowers it progressively rather than only at abrupt shifts, and Corollary E.2 bounds the probability that the gate accepts or keeps a surrogate whose true gap exceeds the margin. The corollary controls the decision error of the gate; it does not certify that the centroid distance tracks every form of quality degradation, which is why we also report the false-rollback and detection rates in Table 5b.

**Serving many session-local adapters.** A LoRA adapter of rank  $r$  on the attention and MLP projections of LLaMA-2-7B has about  $2.5r$  million parameters ( $\approx 20\text{M}$  parameters, or roughly 40 MB in bfloat16, for  $r = 8$ ), i.e., well under 1% of the surrogate. Multi-adapter serving systems batch requests from many adapters over a single copy of the base weights and page adapters between host and GPU memory (Sheng et al., 2023), so GPU residency is bounded by the number of concurrently *switched* sessions rather than by the total number of sessions, and an adapter is discarded when its session ends or rolls back permanently. The mining and adaptation steps for a session are independent of other sessions and can be scheduled as short background jobs on the surrogate replica. We did not benchmark multi-node concurrent adapter scheduling; the reported latencies are for the single-session serving path.

## H BROADER IMPACT

SOMA aims to reduce the cost and latency of multi-turn LLM serving by replacing repeated large-model inference with session-local surrogate serving. This can make interactive LLM systems more efficient, accessible, and environmentally sustainable. At the same time, using a smaller surrogate may introduce quality degradation if the dialogue leaves the learned local state, especially in high-stakes settings such as medical, legal, or financial assistance. SOMA mitigates this risk through semantic acceptance and drift-aware rollback, but deployment should still include monitoring, conservative thresholds, and human oversight where errors may cause harm. Since SOMA uses dialogue history for local adaptation, practical deployments should also apply data minimization, retention limits, and privacy-preserving processing.

## I LIMITATIONS AND FUTURE WORK

SOMA has several limitations. First, the surrogate’s capacity limits how closely it can match the original model. When the small-large model gap is very large, some behaviors cannot be recovered by session-local adaptation alone. Second, soft-prompt mining requires access to the surrogate tokenizer and embedding space, which makes SOMA harder to deploy in strict black-box API settings. Third, SOMA is designed for locally coherent dialogues. Abrupt topic changes, noisy warm-start windows, or strong paraphrase shifts can weaken the learned local directions, although the drift gate is designed to reduce persistent degradation. Finally, SOMA introduces a one-time warm-start cost, so it is most suitable for medium-to-long sessions where post-switch savings can amortize the initial adaptation overhead. **Two further limitations concern evaluation.** Gradual topic drift is detected by the same continuous centroid statistic as abrupt shifts, but we have only stress-tested abrupt shifts (Table 5b); quality may degrade before the rollback margin is exhausted. We also have not compared SOMA against a History-FT variant weighted by raw output divergence or against uniformly sampled local turns under the same LoRA budget (Appendix G), so the incremental value of prompt mining over simpler weightings is supported by the component ablation rather than by a direct head-to-head comparison.

Future work can improve SOMA along several directions. Better drift detectors could make rollback more reliable under topic shifts. Approximate mining methods could reduce the need for internal surrogate access. Multi-region adaptation could support conversations with multiple topics, and privacy-preserving mining could make the method safer for sensitive dialogue settings. Another promising direction is to amortize soft-prompt search across related sessions and extend SOMA to multimodal serving.